```
cudnn lib dir: /nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/nvidia/cudnn/

================================================================================
  TCIA PANCREAS CLASSIFICATION
  MULTI-MODEL 3-FOLD CROSS-VALIDATION PIPELINE
================================================================================

STEP 1: READING MANIFEST
--------------------------------------------------------------------------------
  Total patches:   1464 (244 tumor, 1220 non-tumor)
  Unique patients: 80

STEP 2: SETTING UP 3-FOLD CROSS-VALIDATION
--------------------------------------------------------------------------------
  Patient-level splitting (prevents data leakage):

    Fold 1:  26 patients,  480 patches (80 tumor, 400 non-tumor)
    Fold 2:  27 patients,  492 patches (82 tumor, 410 non-tumor)
    Fold 3:  27 patients,  492 patches (82 tumor, 410 non-tumor)

  Device: cuda
  GPU:    NVIDIA A100-SXM4-40GB MIG 3g.20gb
  VRAM:   20.9 GB

  Models to train: ResNet3D, VGG3D


********************************************************************************
  TRAINING MODEL 1/2: ResNet3D
  Pipeline elapsed: 0 min
********************************************************************************
```

```
#  MODEL: ResNet3D (resnet3d)
##############################################################################
Collapse Output
==============================================================================
  ResNet3D - FOLD 1 / 3
==============================================================================

  Split: 46 train patients, 8 val patients, 26 test patients
  Datasets:
    train:   840 patches (140 positive, 700 negative, ratio 1:5)
    val  :   144 patches (24 positive, 120 negative, ratio 1:5)
    test :   480 patches (80 positive, 400 negative, ratio 1:5)
  Parameters: 3.62M (3,615,970 total)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1918 TrA:0.471 | VaL:0.1313 VaA:0.833 VaAUC:0.922 | LR:6.2e-05 | 25s/ep | fold:1m total:1m
/nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/torch/optim/lr_scheduler.py:243: UserWarning: The epoch parameter in `scheduler
ing deprecated where possible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is different from None, the clos
hainable form, where available. Please open an issue if you are unable to replicate your use case: https://github.com/pytorch/pytorch/issues/new/choo
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.0675 TrA:0.901 | VaL:0.0193 VaA:0.979 VaAUC:0.997 | LR:3.0e-04 | 22s/ep | fold:2m total:2m
    Ep  10/100 | TrL:0.0276 TrA:0.965 | VaL:0.0221 VaA:0.965 VaAUC:0.998 | LR:3.0e-04 | 22s/ep | fold:4m total:4m
    Ep  15/100 | TrL:0.0234 TrA:0.970 | VaL:0.0091 VaA:0.986 VaAUC:0.999 | LR:2.9e-04 | 22s/ep | fold:6m total:6m
    Ep  20/100 | TrL:0.0244 TrA:0.975 | VaL:0.0095 VaA:0.986 VaAUC:0.999 | LR:2.8e-04 | 22s/ep | fold:8m total:8m
    Ep  25/100 | TrL:0.0118 TrA:0.992 | VaL:0.0108 VaA:0.986 VaAUC:0.998 | LR:2.7e-04 | 22s/ep | fold:9m total:9m
    Ep  30/100 | TrL:0.0149 TrA:0.980 | VaL:0.0105 VaA:0.986 VaAUC:0.999 | LR:2.5e-04 | 22s/ep | fold:11m total:11m
    Ep  35/100 | TrL:0.0099 TrA:0.992 | VaL:0.0122 VaA:0.986 VaAUC:0.999 | LR:2.3e-04 | 22s/ep | fold:13m total:13m
    Ep  40/100 | TrL:0.0124 TrA:0.989 | VaL:0.0428 VaA:0.958 VaAUC:0.996 | LR:2.1e-04 | 22s/ep | fold:15m total:15m

    Early stopping at epoch 42 (no improvement for 25 epochs)

  ResNet3D Fold 1 training complete: 42 epochs in 15.7 min (best val auc: 0.9993, best val loss: 0.0078)
```

```
Evaluating ResNet3D fold 1 on test set (480 samples) with TTA...

============================================================
EVALUATION METRICS - ResNet3D FOLD 1 TEST
============================================================
Sensitivity (Recall):  1.0000
Specificity:           0.9775
Precision:             0.8989
Accuracy:              0.9812
F1 Score:              0.9467
AUC-ROC:               0.9997

Confusion Matrix:
  True Positives:    80  (Caught nodules)
  True Negatives:   391  (Correct normals)
  False Positives:    9  (False alarms)
  False Negatives:    0  (Missed nodules)
============================================================


Generating Grad-CAM visualizations for ResNet3D...
  TP: 3 figure(s) saved
  TN: 3 figure(s) saved
  FP: 3 figure(s) saved
  FN: 1 figure(s) saved
Grad-CAM figures saved to: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/gradcam/resnet3d

================================================================================
ResNet3D - FOLD 2 / 3
================================================================================

Split: 46 train patients, 7 val patients, 27 test patients
Datasets:
  train:   834 patches (139 positive, 695 negative, ratio 1:5)
```

```
  val  :   138 patches (23 positive, 115 negative, ratio 1:5)
  test :   492 patches (82 positive, 410 negative, ratio 1:5)

Training (max 100 epochs, patience=25)...

  Ep   1/100 | TrL:0.1745 TrA:0.550 | VaL:0.1608 VaA:0.659 VaAUC:0.960 | LR:6.2e-05 | 22s/ep | fold:0m total:17m
nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/torch/optim/lr_scheduler.py:243: UserWarning: The epo
ng deprecated where possible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is dif
ainable form, where available. Please open an issue if you are unable to replicate your use case: https://github.com/pytor
 warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
  Ep   5/100 | TrL:0.0409 TrA:0.954 | VaL:0.2162 VaA:0.623 VaAUC:0.979 | LR:3.0e-04 | 22s/ep | fold:2m total:18m
  Ep  10/100 | TrL:0.0359 TrA:0.953 | VaL:0.0110 VaA:0.993 VaAUC:0.999 | LR:3.0e-04 | 22s/ep | fold:4m total:20m
  Ep  15/100 | TrL:0.0174 TrA:0.975 | VaL:0.0043 VaA:1.000 VaAUC:1.000 | LR:2.9e-04 | 22s/ep | fold:6m total:22m
  Ep  20/100 | TrL:0.0269 TrA:0.972 | VaL:0.0101 VaA:0.993 VaAUC:0.998 | LR:2.8e-04 | 22s/ep | fold:7m total:24m
  Ep  25/100 | TrL:0.0114 TrA:0.989 | VaL:0.0130 VaA:0.986 VaAUC:0.998 | LR:2.7e-04 | 22s/ep | fold:9m total:26m
  Ep  30/100 | TrL:0.0130 TrA:0.984 | VaL:0.0386 VaA:0.957 VaAUC:0.998 | LR:2.5e-04 | 22s/ep | fold:11m total:27m
  Ep  35/100 | TrL:0.0046 TrA:0.993 | VaL:0.0188 VaA:0.971 VaAUC:0.999 | LR:2.3e-04 | 22s/ep | fold:13m total:29m
  Ep  40/100 | TrL:0.0042 TrA:0.999 | VaL:0.0599 VaA:0.942 VaAUC:0.995 | LR:2.1e-04 | 22s/ep | fold:15m total:31m

  Early stopping at epoch 40 (no improvement for 25 epochs)

ResNet3D Fold 2 training complete: 40 epochs in 14.7 min (best val auc: 1.0000, best val loss: 0.0043)
Estimated time remaining for ResNet3D: ~15 min

Evaluating ResNet3D fold 2 on test set (492 samples) with TTA...
```

```
============================================================
  EVALUATION METRICS - ResNet3D FOLD 2 TEST
============================================================
  Sensitivity (Recall):  0.9634
  Specificity:           0.9805
  Precision:             0.9080
  Accuracy:              0.9776
  F1 Score:              0.9349
  AUC-ROC:               0.9962

  Confusion Matrix:
    True Positives:    79  (Caught nodules)
    True Negatives:   402  (Correct normals)
    False Positives:    8  (False alarms)
    False Negatives:    3  (Missed nodules)
============================================================


================================================================================
  ResNet3D - FOLD 3 / 3
================================================================================

  Split: 46 train patients, 7 val patients, 27 test patients
  Datasets:
    train:   846 patches (141 positive, 705 negative, ratio 1:5)
    val  :   126 patches (21 positive, 105 negative, ratio 1:5)
    test :   492 patches (82 positive, 410 negative, ratio 1:5)

  Training (max 100 epochs, patience=25)...
```

```
    Ep   1/100 | TrL:0.1821 TrA:0.537 | VaL:0.1807 VaA:0.389 VaAUC:0.926 | LR:6.2e-05 | 22s/ep | fold:0m total:32m
/nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/torch/optim/lr_scheduler.py:243: UserWarning: The
ing deprecated where possible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is
hainable form, where available. Please open an issue if you are unable to replicate your use case: https://github.com/p
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.0578 TrA:0.915 | VaL:0.0909 VaA:0.849 VaAUC:0.982 | LR:3.0e-04 | 22s/ep | fold:2m total:33m
    Ep  10/100 | TrL:0.0219 TrA:0.978 | VaL:0.0326 VaA:0.960 VaAUC:0.991 | LR:3.0e-04 | 22s/ep | fold:4m total:35m
    Ep  15/100 | TrL:0.0125 TrA:0.983 | VaL:0.0338 VaA:0.968 VaAUC:0.987 | LR:2.9e-04 | 22s/ep | fold:6m total:37m
    Ep  20/100 | TrL:0.0088 TrA:0.992 | VaL:0.0489 VaA:0.960 VaAUC:0.988 | LR:2.8e-04 | 22s/ep | fold:7m total:39m
    Ep  25/100 | TrL:0.0069 TrA:0.992 | VaL:0.0238 VaA:0.976 VaAUC:0.995 | LR:2.7e-04 | 22s/ep | fold:9m total:41m
    Ep  30/100 | TrL:0.0086 TrA:0.988 | VaL:0.0440 VaA:0.976 VaAUC:0.991 | LR:2.5e-04 | 22s/ep | fold:11m total:43m
    Ep  35/100 | TrL:0.0103 TrA:0.992 | VaL:0.0708 VaA:0.960 VaAUC:0.990 | LR:2.3e-04 | 22s/ep | fold:13m total:45m
    Ep  40/100 | TrL:0.0063 TrA:0.993 | VaL:0.0233 VaA:0.976 VaAUC:0.995 | LR:2.1e-04 | 22s/ep | fold:15m total:46m
    Ep  45/100 | TrL:0.0038 TrA:0.995 | VaL:0.0477 VaA:0.976 VaAUC:0.992 | LR:1.9e-04 | 22s/ep | fold:17m total:48m
    Ep  50/100 | TrL:0.0030 TrA:0.994 | VaL:0.0335 VaA:0.968 VaAUC:0.993 | LR:1.6e-04 | 22s/ep | fold:18m total:50m

    Early stopping at epoch 51 (no improvement for 25 epochs)

  ResNet3D Fold 3 training complete: 51 epochs in 18.9 min (best val auc: 0.9973, best val loss: 0.0297)

  Evaluating ResNet3D fold 3 on test set (492 samples) with TTA...

============================================================
  EVALUATION METRICS - ResNet3D FOLD 3 TEST
============================================================
  Sensitivity (Recall):  0.9756
  Specificity:           0.9293
  Precision:             0.7339
  Accuracy:              0.9370
  F1 Score:              0.8377
  AUC-ROC:               0.9938
```

```
  Confusion Matrix:
    True Positives:    80  (Caught nodules)
    True Negatives:   381  (Correct normals)
    False Positives:   29  (False alarms)
    False Negatives:    2  (Missed nodules)
============================================================



================================================================================
  ResNet3D AGGREGATE RESULTS (3-FOLD CV)
================================================================================

  Metric            Fold 1   Fold 2   Fold 3     Mean+/-Std
  ----------------------------------------------------------
  sensitivity       1.0000   0.9634   0.9756 0.9797+/-0.0152
  specificity       0.9775   0.9805   0.9293 0.9624+/-0.0235
  precision         0.8989   0.9080   0.7339 0.8470+/-0.0800
  accuracy          0.9812   0.9776   0.9370 0.9653+/-0.0201
  auc_roc           0.9997   0.9962   0.9938 0.9965+/-0.0024
  f1                0.9467   0.9349   0.8377 0.9065+/-0.0489

  Aggregate (all folds combined, 1464 samples):
    sensitivity     0.9795
    specificity     0.9623
    precision       0.8386
    accuracy        0.9652
    auc_roc         0.9960
    f1              0.9036
```

```
  ROC curve saved: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/resnet3d/roc_curve_kfold.png
  Training curves saved: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/resnet3d/training_curves_kfold.png

******************************************************************************
  TRAINING MODEL 2/2: VGG3D
  Pipeline elapsed: 51 min
******************************************************************************

##############################################################################
#  MODEL: VGG3D (vgg3d)
##############################################################################

==============================================================================
  VGG3D - FOLD 1 / 3
==============================================================================

  Split: 46 train patients, 8 val patients, 26 test patients
  Datasets:
    train:   840 patches (140 positive, 700 negative, ratio 1:5)
    val  :   144 patches (24 positive, 120 negative, ratio 1:5)
    test :   480 patches (80 positive, 400 negative, ratio 1:5)
  Parameters: 0.87M (866,338 total)

  Training (max 100 epochs, patience=25)...
```

```
    Ep   1/100 | TrL:0.2041 TrA:0.511 | VaL:0.1526 VaA:0.847 VaAUC:0.962 | LR:6.2e-05 | 14s/ep | fold:0m total:51m
/nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/torch/optim/lr_scheduler.py:243: UserWarning: The epo
ing deprecated where possible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is dif
hainable form, where available. Please open an issue if you are unable to replicate your use case: https://github.com/pytor
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.0853 TrA:0.885 | VaL:0.0550 VaA:0.944 VaAUC:0.995 | LR:3.0e-04 | 14s/ep | fold:1m total:52m
    Ep  10/100 | TrL:0.0422 TrA:0.943 | VaL:0.0669 VaA:0.931 VaAUC:0.997 | LR:3.0e-04 | 14s/ep | fold:2m total:53m
    Ep  15/100 | TrL:0.0245 TrA:0.956 | VaL:0.0161 VaA:0.958 VaAUC:0.998 | LR:2.9e-04 | 14s/ep | fold:4m total:55m
    Ep  20/100 | TrL:0.0235 TrA:0.969 | VaL:0.0205 VaA:0.972 VaAUC:0.997 | LR:2.8e-04 | 14s/ep | fold:5m total:56m
    Ep  25/100 | TrL:0.0214 TrA:0.976 | VaL:0.0207 VaA:0.986 VaAUC:0.997 | LR:2.7e-04 | 14s/ep | fold:6m total:57m
    Ep  30/100 | TrL:0.0161 TrA:0.982 | VaL:0.0140 VaA:0.965 VaAUC:0.998 | LR:2.5e-04 | 14s/ep | fold:7m total:58m
    Ep  35/100 | TrL:0.0182 TrA:0.982 | VaL:0.0168 VaA:0.972 VaAUC:0.999 | LR:2.3e-04 | 14s/ep | fold:8m total:59m
    Ep  40/100 | TrL:0.0080 TrA:0.990 | VaL:0.0108 VaA:0.986 VaAUC:0.998 | LR:2.1e-04 | 14s/ep | fold:9m total:60m
    Ep  45/100 | TrL:0.0047 TrA:0.996 | VaL:0.0075 VaA:0.986 VaAUC:1.000 | LR:1.9e-04 | 14s/ep | fold:11m total:62m
    Ep  50/100 | TrL:0.0019 TrA:0.999 | VaL:0.0065 VaA:0.986 VaAUC:0.999 | LR:1.6e-04 | 14s/ep | fold:12m total:63m
    Ep  55/100 | TrL:0.0065 TrA:0.994 | VaL:0.0062 VaA:0.993 VaAUC:1.000 | LR:1.4e-04 | 14s/ep | fold:13m total:64m
    Ep  60/100 | TrL:0.0067 TrA:0.993 | VaL:0.0178 VaA:0.986 VaAUC:1.000 | LR:1.1e-04 | 14s/ep | fold:14m total:65m
    Ep  65/100 | TrL:0.0070 TrA:0.993 | VaL:0.0169 VaA:0.986 VaAUC:0.998 | LR:9.0e-05 | 14s/ep | fold:15m total:66m
    Ep  70/100 | TrL:0.0034 TrA:0.995 | VaL:0.0042 VaA:0.993 VaAUC:1.000 | LR:6.8e-05 | 14s/ep | fold:17m total:68m
    Ep  75/100 | TrL:0.0013 TrA:0.998 | VaL:0.0210 VaA:0.979 VaAUC:0.999 | LR:4.8e-05 | 14s/ep | fold:18m total:69m
    Ep  80/100 | TrL:0.0009 TrA:0.999 | VaL:0.0069 VaA:0.986 VaAUC:1.000 | LR:3.2e-05 | 14s/ep | fold:19m total:70m

    Early stopping at epoch 81 (no improvement for 25 epochs)

  VGG3D Fold 1 training complete: 81 epochs in 19.2 min (best val auc: 1.0000, best val loss: 0.0028)
  Estimated time remaining for VGG3D: ~38 min

  Evaluating VGG3D fold 1 on test set (480 samples) with TTA...
```

```
============================================================
  EVALUATION METRICS - VGG3D FOLD 1 TEST
============================================================
  Sensitivity (Recall):  0.9875
  Specificity:           0.9800
  Precision:             0.9080
  Accuracy:              0.9812
  F1 Score:              0.9461
  AUC-ROC:               0.9994

  Confusion Matrix:
    True Positives:    79  (Caught nodules)
    True Negatives:   392  (Correct normals)
    False Positives:    8  (False alarms)
    False Negatives:    1  (Missed nodules)
============================================================


  Generating Grad-CAM visualizations for VGG3D...
    TP: 3 figure(s) saved
    TN: 3 figure(s) saved
    FP: 3 figure(s) saved
    FN: 1 figure(s) saved
  Grad-CAM figures saved to: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/gradcam/vgg3d

================================================================================
  VGG3D - FOLD 2 / 3
================================================================================
```

```
  Split: 46 train patients, 7 val patients, 27 test patients
  Datasets:
    train:   834 patches (139 positive, 695 negative, ratio 1:5)
    val  :   138 patches (23 positive, 115 negative, ratio 1:5)
    test :   492 patches (82 positive, 410 negative, ratio 1:5)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.2212 TrA:0.487 | VaL:0.2363 VaA:0.167 VaAUC:0.968 | LR:6.2e-05 | 15s/ep | fold:0m total:71m
/nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/torch/optim/lr_scheduler.py:243: UserWarning: The ep
ing deprecated where possible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is di
hainable form, where available. Please open an issue if you are unable to replicate your use case: https://github.com/pyto
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.0706 TrA:0.893 | VaL:0.0272 VaA:0.978 VaAUC:0.997 | LR:3.0e-04 | 14s/ep | fold:1m total:72m
    Ep  10/100 | TrL:0.0443 TrA:0.940 | VaL:0.0160 VaA:0.978 VaAUC:0.998 | LR:3.0e-04 | 14s/ep | fold:2m total:73m
    Ep  15/100 | TrL:0.0276 TrA:0.962 | VaL:0.0137 VaA:0.971 VaAUC:0.997 | LR:2.9e-04 | 14s/ep | fold:4m total:74m
    Ep  20/100 | TrL:0.0248 TrA:0.970 | VaL:0.0105 VaA:0.978 VaAUC:0.998 | LR:2.8e-04 | 14s/ep | fold:5m total:75m
    Ep  25/100 | TrL:0.0158 TrA:0.983 | VaL:0.0104 VaA:0.993 VaAUC:0.998 | LR:2.7e-04 | 14s/ep | fold:6m total:77m
    Ep  30/100 | TrL:0.0173 TrA:0.981 | VaL:0.0113 VaA:0.978 VaAUC:1.000 | LR:2.5e-04 | 14s/ep | fold:7m total:78m
    Ep  35/100 | TrL:0.0124 TrA:0.988 | VaL:0.0101 VaA:0.986 VaAUC:0.999 | LR:2.3e-04 | 14s/ep | fold:8m total:79m
    Ep  40/100 | TrL:0.0110 TrA:0.984 | VaL:0.0071 VaA:0.993 VaAUC:1.000 | LR:2.1e-04 | 14s/ep | fold:9m total:80m
    Ep  45/100 | TrL:0.0046 TrA:0.993 | VaL:0.0196 VaA:0.957 VaAUC:1.000 | LR:1.9e-04 | 14s/ep | fold:11m total:81m
    Ep  50/100 | TrL:0.0071 TrA:0.993 | VaL:0.0087 VaA:0.986 VaAUC:1.000 | LR:1.6e-04 | 14s/ep | fold:12m total:83m
    Ep  55/100 | TrL:0.0108 TrA:0.988 | VaL:0.0478 VaA:0.949 VaAUC:1.000 | LR:1.4e-04 | 14s/ep | fold:13m total:84m
    Ep  60/100 | TrL:0.0050 TrA:0.995 | VaL:0.0021 VaA:1.000 VaAUC:1.000 | LR:1.1e-04 | 14s/ep | fold:14m total:85m
    Ep  65/100 | TrL:0.0019 TrA:0.999 | VaL:0.0008 VaA:1.000 VaAUC:1.000 | LR:9.0e-05 | 14s/ep | fold:15m total:86m
    Ep  70/100 | TrL:0.0022 TrA:0.999 | VaL:0.0029 VaA:0.993 VaAUC:1.000 | LR:6.8e-05 | 14s/ep | fold:17m total:87m
    Ep  75/100 | TrL:0.0014 TrA:0.998 | VaL:0.0005 VaA:1.000 VaAUC:1.000 | LR:4.8e-05 | 14s/ep | fold:18m total:88m
```

```
  Ep  75/100 | TrL:0.0014 TrA:0.998 | VaL:0.0005 VaA:1.000 VaAUC:1.000 | LR:4.8e-05 | 14s/ep | fold:18m total:88m
  Ep  80/100 | TrL:0.0003 TrA:1.000 | VaL:0.0003 VaA:1.000 VaAUC:1.000 | LR:3.2e-05 | 14s/ep | fold:19m total:90m
  Ep  85/100 | TrL:0.0005 TrA:0.999 | VaL:0.0001 VaA:1.000 VaAUC:1.000 | LR:1.8e-05 | 14s/ep | fold:20m total:91m
  Ep  90/100 | TrL:0.0002 TrA:1.000 | VaL:0.0002 VaA:1.000 VaAUC:1.000 | LR:8.1e-06 | 14s/ep | fold:21m total:92m
  Ep  95/100 | TrL:0.0003 TrA:1.000 | VaL:0.0001 VaA:1.000 VaAUC:1.000 | LR:2.0e-06 | 14s/ep | fold:22m total:93m
  Ep 100/100 | TrL:0.0003 TrA:1.000 | VaL:0.0001 VaA:1.000 VaAUC:1.000 | LR:0.0e+00 | 14s/ep | fold:24m total:94m

VGG3D Fold 2 training complete: 100 epochs in 23.6 min (best val auc: 1.0000, best val loss: 0.0001)
Estimated time remaining for VGG3D: ~21 min

Evaluating VGG3D fold 2 on test set (492 samples) with TTA...

============================================================
EVALUATION METRICS - VGG3D FOLD 2 TEST
============================================================
Sensitivity (Recall):  0.9634
Specificity:           0.9951
Precision:             0.9753
Accuracy:              0.9898
F1 Score:              0.9693
AUC-ROC:               0.9978

Confusion Matrix:
  True Positives:    79  (Caught nodules)
  True Negatives:   408  (Correct normals)
  False Positives:    2  (False alarms)
  False Negatives:    3  (Missed nodules)
```

```
================================================================================
  VGG3D - FOLD 3 / 3
================================================================================

  Split: 46 train patients, 7 val patients, 27 test patients
  Datasets:
    train:   846 patches (141 positive, 705 negative, ratio 1:5)
    val  :   126 patches (21 positive, 105 negative, ratio 1:5)
    test :   492 patches (82 positive, 410 negative, ratio 1:5)

  Training (max 100 epochs, patience=25)...

    Ep   1/100 | TrL:0.1833 TrA:0.550 | VaL:0.1675 VaA:0.619 VaAUC:0.916 | LR:6.2e-05 | 15s/ep | fold:0m total:95m
/nlsasfs/home/aikosh/prod-aikosh31/.venv/lib/python3.10/site-packages/torch/optim/lr_scheduler.py:243: UserWarning: The ep
ing deprecated where possible. Please use `scheduler.step()` to step the scheduler. During the deprecation, if epoch is di
hainable form, where available. Please open an issue if you are unable to replicate your use case: https://github.com/pytc
  warnings.warn(EPOCH_DEPRECATION_WARNING, UserWarning)
    Ep   5/100 | TrL:0.0479 TrA:0.947 | VaL:0.0636 VaA:0.905 VaAUC:0.975 | LR:3.0e-04 | 14s/ep | fold:1m total:96m
    Ep  10/100 | TrL:0.0231 TrA:0.970 | VaL:0.0345 VaA:0.960 VaAUC:0.985 | LR:3.0e-04 | 14s/ep | fold:2m total:97m
    Ep  15/100 | TrL:0.0107 TrA:0.992 | VaL:0.0696 VaA:0.952 VaAUC:0.979 | LR:2.9e-04 | 14s/ep | fold:4m total:98m
    Ep  20/100 | TrL:0.0085 TrA:0.993 | VaL:0.0502 VaA:0.952 VaAUC:0.990 | LR:2.8e-04 | 14s/ep | fold:5m total:99m
    Ep  25/100 | TrL:0.0131 TrA:0.987 | VaL:0.0262 VaA:0.968 VaAUC:0.991 | LR:2.7e-04 | 14s/ep | fold:6m total:101m
    Ep  30/100 | TrL:0.0043 TrA:0.995 | VaL:0.0724 VaA:0.960 VaAUC:0.994 | LR:2.5e-04 | 14s/ep | fold:7m total:102m
    Ep  35/100 | TrL:0.0052 TrA:0.994 | VaL:0.0412 VaA:0.976 VaAUC:0.993 | LR:2.3e-04 | 14s/ep | fold:8m total:103m
    Ep  40/100 | TrL:0.0153 TrA:0.985 | VaL:0.0354 VaA:0.976 VaAUC:0.995 | LR:2.1e-04 | 14s/ep | fold:10m total:104m
    Ep  45/100 | TrL:0.0120 TrA:0.988 | VaL:0.0370 VaA:0.968 VaAUC:0.995 | LR:1.9e-04 | 14s/ep | fold:11m total:105m
    Ep  50/100 | TrL:0.0065 TrA:0.995 | VaL:0.0311 VaA:0.976 VaAUC:0.996 | LR:1.6e-04 | 14s/ep | fold:12m total:107m
    Ep  55/100 | TrL:0.0053 TrA:0.998 | VaL:0.0383 VaA:0.976 VaAUC:0.996 | LR:1.4e-04 | 14s/ep | fold:13m total:108m
    Ep  60/100 | TrL:0.0005 TrA:1.000 | VaL:0.0298 VaA:0.976 VaAUC:0.996 | LR:1.1e-04 | 14s/ep | fold:14m total:109m
    Ep  65/100 | TrL:0.0004 TrA:1.000 | VaL:0.0351 VaA:0.968 VaAUC:0.996 | LR:9.0e-05 | 14s/ep | fold:15m total:110m
    Ep  70/100 | TrL:0.0051 TrA:0.995 | VaL:0.0385 VaA:0.976 VaAUC:0.995 | LR:6.8e-05 | 14s/ep | fold:17m total:111m
```

```
    Early stopping at epoch 72 (no improvement for 25 epochs)

  VGG3D Fold 3 training complete: 72 epochs in 17.1 min (best val auc: 0.9977, best val loss: 0.0210)

  Evaluating VGG3D fold 3 on test set (492 samples) with TTA...

============================================================
  EVALUATION METRICS - VGG3D FOLD 3 TEST
============================================================
  Sensitivity (Recall):  0.9390
  Specificity:           0.9854
  Precision:             0.9277
  Accuracy:              0.9776
  F1 Score:              0.9333
  AUC-ROC:               0.9969

  Confusion Matrix:
    True Positives:    77  (Caught nodules)
    True Negatives:   404  (Correct normals)
    False Positives:    6  (False alarms)
    False Negatives:    5  (Missed nodules)
============================================================
```

```
================================================================================
  VGG3D AGGREGATE RESULTS (3-FOLD CV)
================================================================================

  Metric            Fold 1   Fold 2   Fold 3     Mean+/-Std
  -------------------------------------------------------
  sensitivity       0.9875   0.9634   0.9390 0.9633+/-0.0198
  specificity       0.9800   0.9951   0.9854 0.9868+/-0.0063
  precision         0.9080   0.9753   0.9277 0.9370+/-0.0282
  accuracy          0.9812   0.9898   0.9776 0.9829+/-0.0051
  auc_roc           0.9994   0.9978   0.9969 0.9980+/-0.0010
  f1                0.9461   0.9693   0.9333 0.9496+/-0.0149

  Aggregate (all folds combined, 1464 samples):
    sensitivity     0.9631
    specificity     0.9869
    precision       0.9363
    accuracy        0.9829
    auc_roc         0.9977
    f1              0.9495

  ROC curve saved: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/vgg3d/roc_curve_kfold.png
  Training curves saved: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/vgg3d/training_curves_kfold.png

================================================================================
  STEP 4: CROSS-MODEL COMPARISON
================================================================================

  Model           Params     sensitivity     specificity     precision       accuracy        auc_roc         f1              Time
  ---------------------------------------------------------------------------------------------------------------------------------
  ResNet3D        3.62M      0.980+/-0.015  0.962+/-0.023  0.847+/-0.080  0.965+/-0.020  0.997+/-0.002  0.906+/-0.049  51m
  VGG3D           0.87M      0.963+/-0.020  0.987+/-0.006  0.937+/-0.028  0.983+/-0.005  0.998+/-0.001  0.950+/-0.015  61m
```

```
Comparison ROC saved: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/comparison_roc.png
Results summary saved: /nlsasfs/home/aikosh/prod-aikosh31/run_pancreas/results/pancreas/plots/results_summary.txt
```